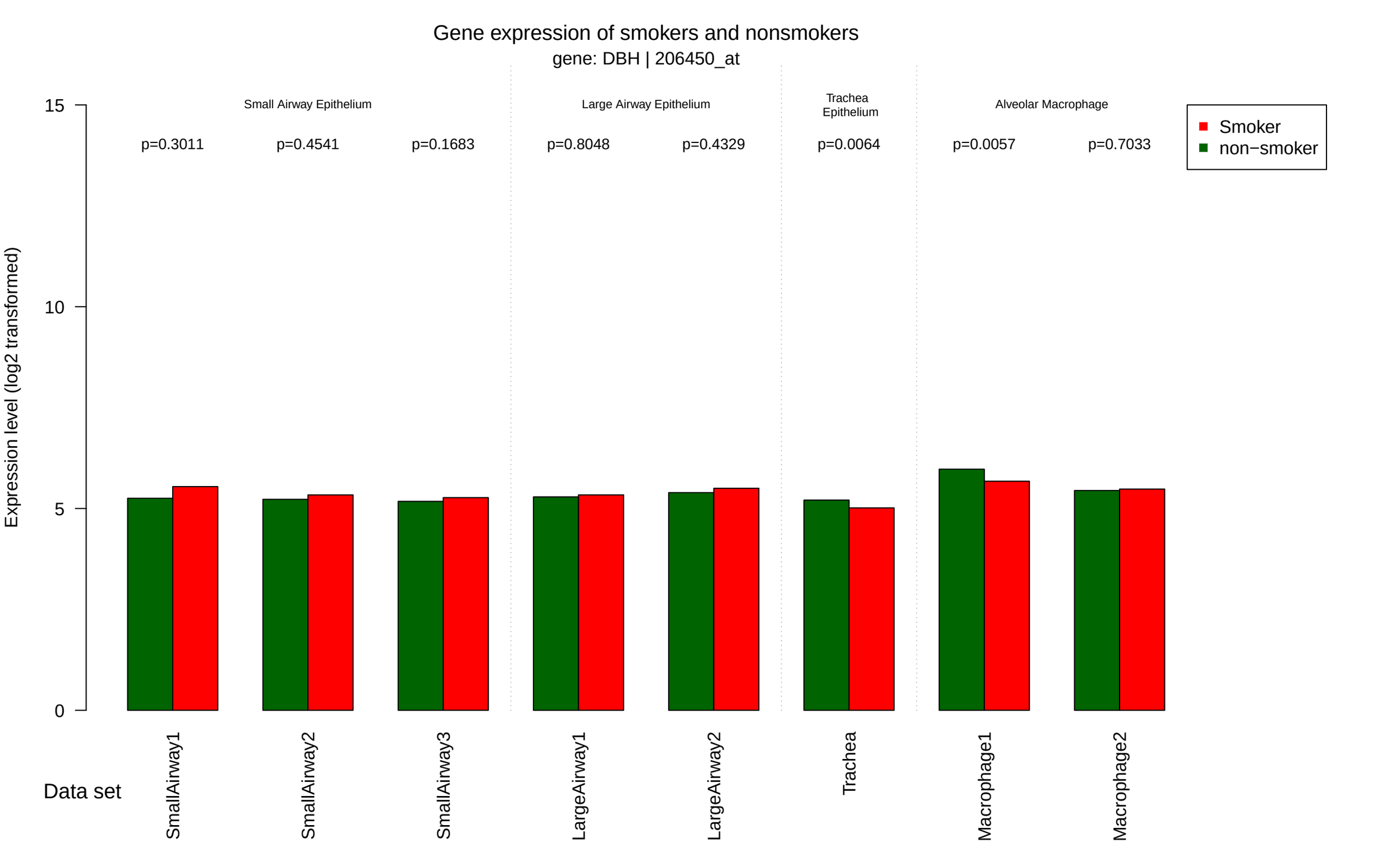
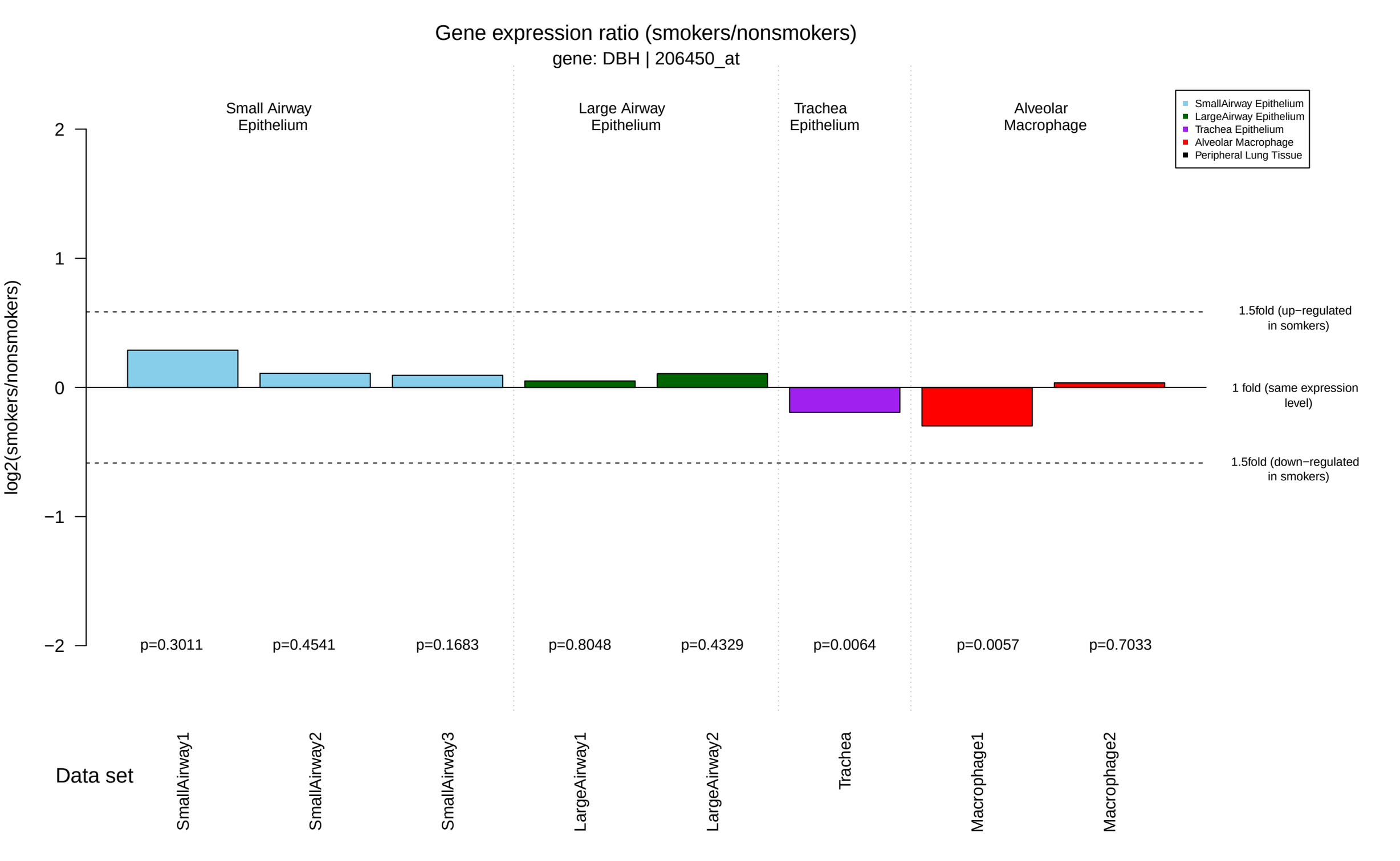
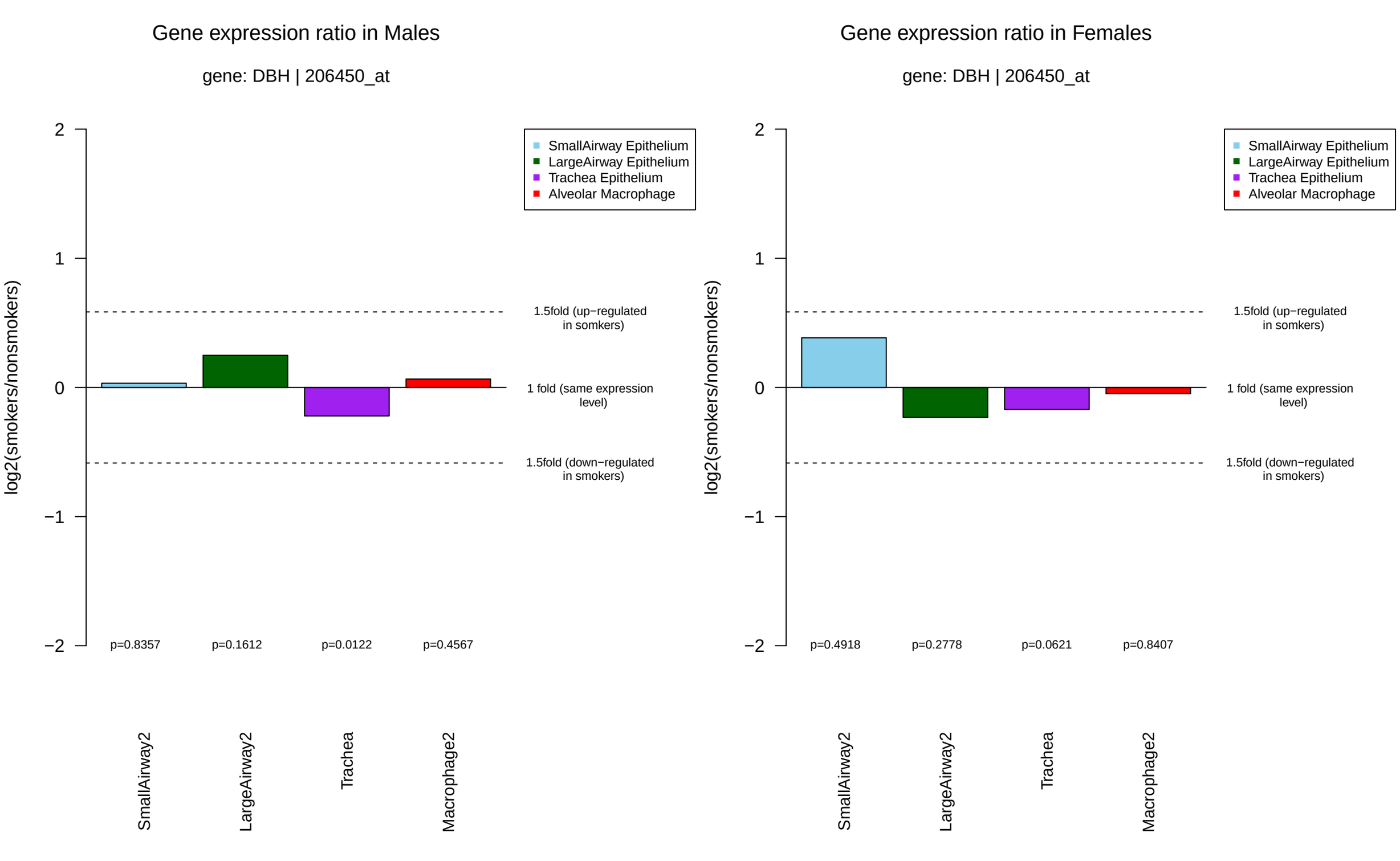
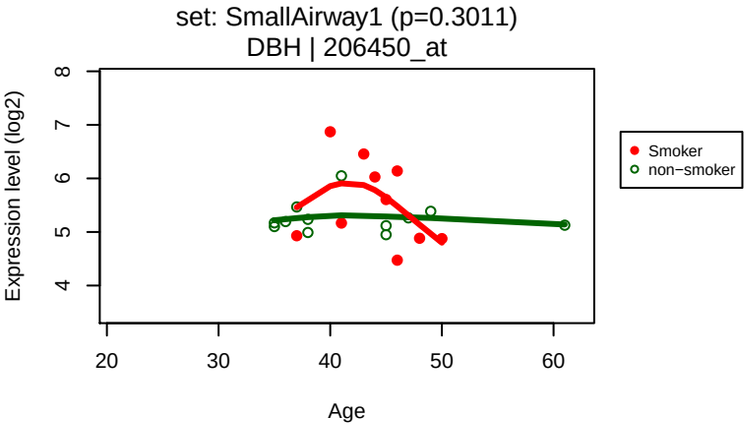
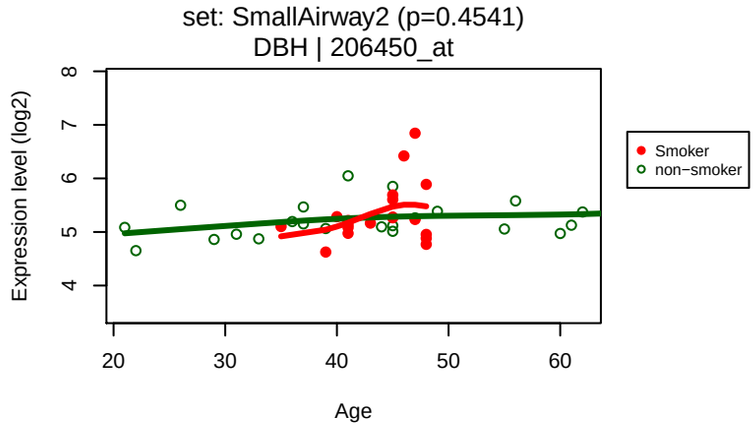
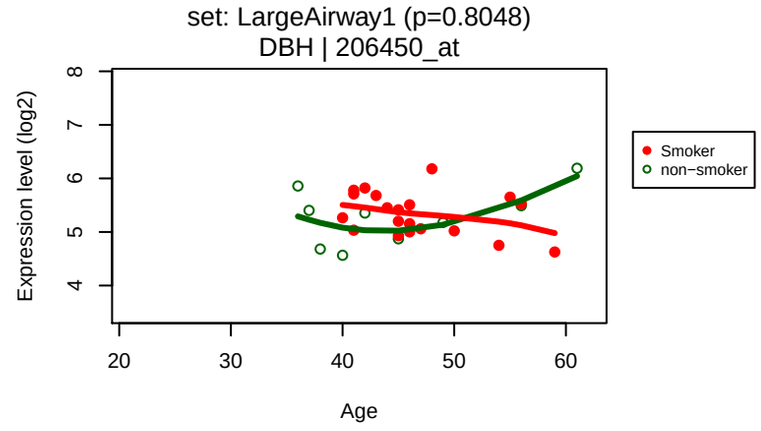

Gene expression of smokers and nonsmokers
gene: DBH | 206450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3011
p=0.4541
p=0.1683
p=0.8048
p=0.4329
p=0.0064
p=0.0057
p=0.7033
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

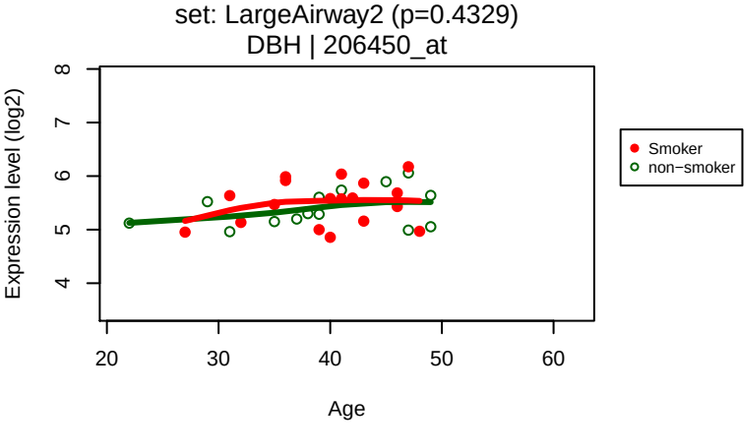

Gene expression ratio (smokers/nonsmokers)
gene: DBH | 206450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3011
p=0.4541
p=0.1683
p=0.8048
p=0.4329
p=0.0064
p=0.0057
p=0.7033
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

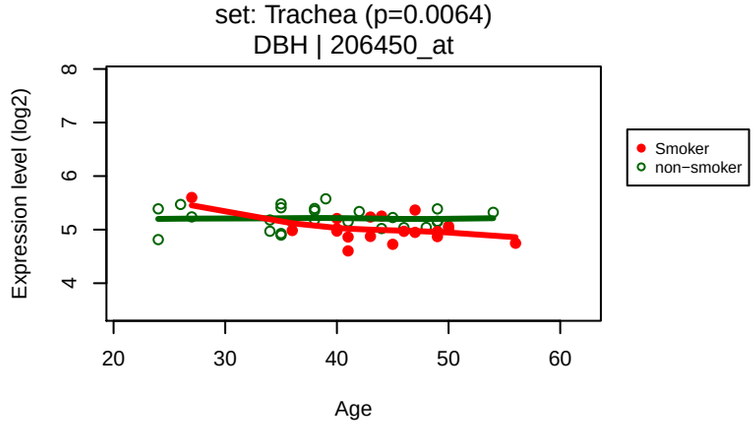

Gene expression ratio in Males
gene: DBH | 206450_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8357
p=0.1612
p=0.0122
p=0.4567
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DBH | 206450_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4918
p=0.2778
p=0.0621
p=0.8407
SmallAirway2
LargeAirway2
Trachea
Macrophage2

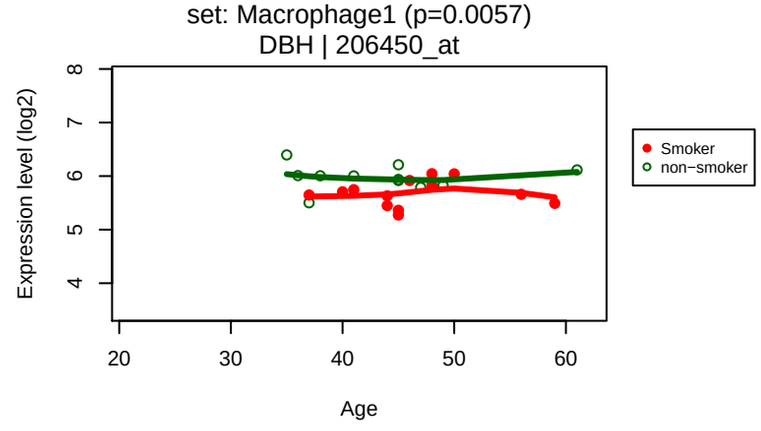

set: SmallAirway1 (p=0.3011)
DBH | 206450_at
Expression level (log2)
Age
Smoker
non−smoker

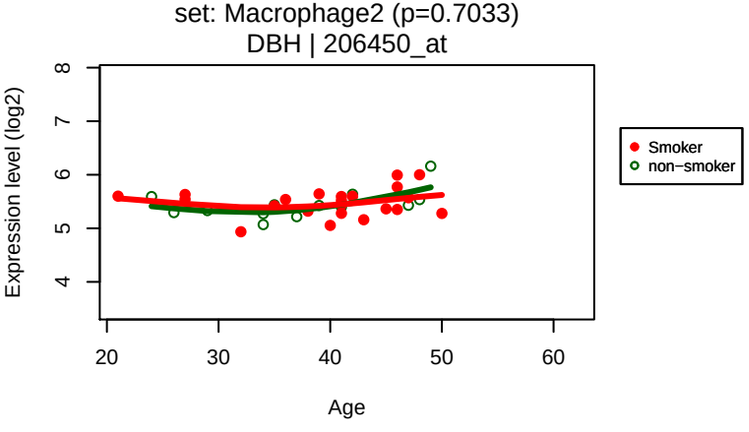

set: SmallAirway2 (p=0.4541)
DBH | 206450_at
Expression level (log2)
Age
Smoker
non−smoker

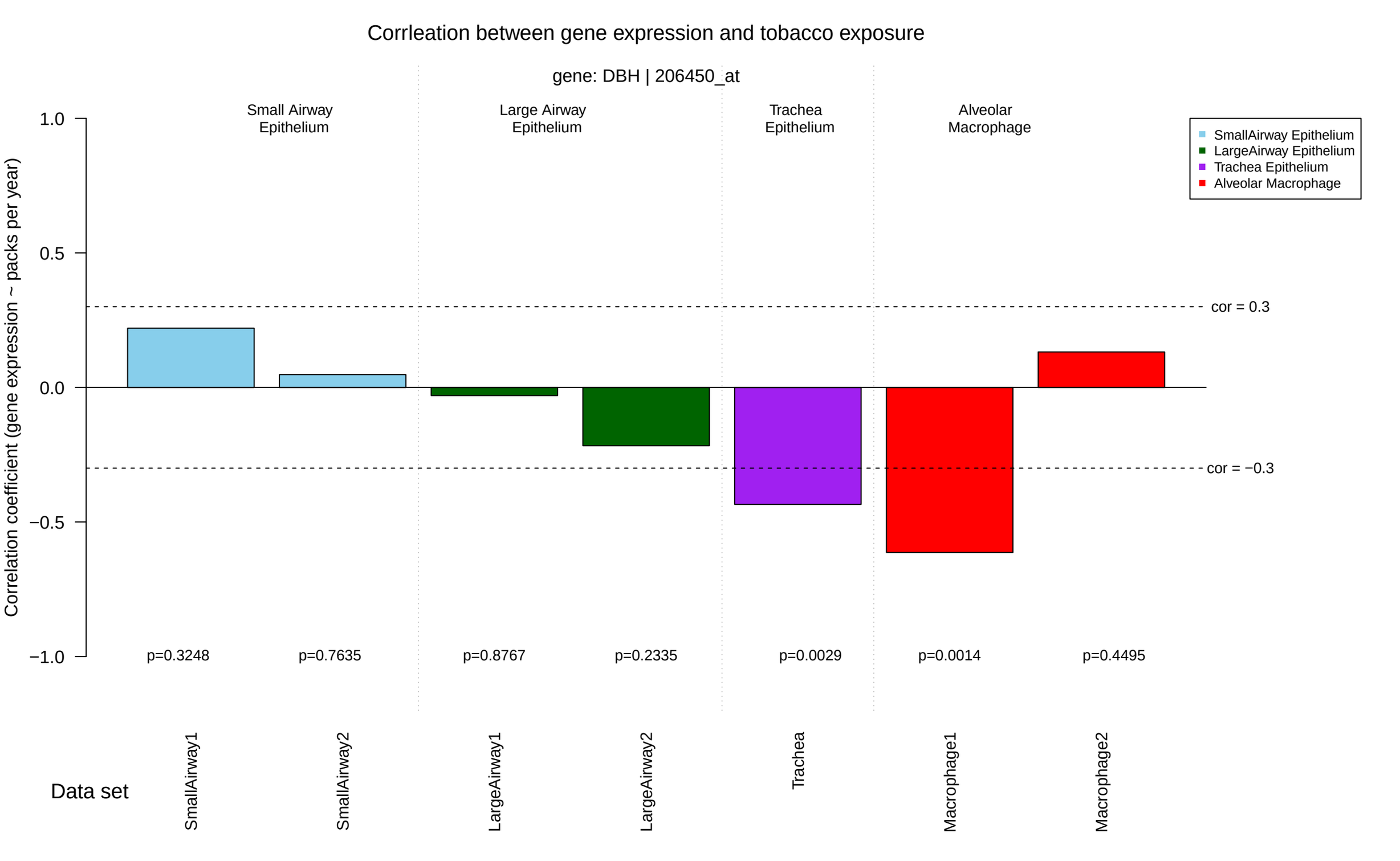

set: LargeAirway1 (p=0.8048)
DBH | 206450_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4329)
DBH | 206450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0064)
DBH | 206450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0057)
DBH | 206450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7033)
DBH | 206450_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DBH | 206450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3248
p=0.7635
p=0.8767
p=0.2335
p=0.0029
p=0.0014
p=0.4495
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2